



02 April 2025

Mr. Joseph McCloud
NAVFAC Mid-Atlantic
NAVFAC MIDLANT
9324 Virginia Ave
Norfolk, VA 23511-3095

RE: **Monthly Progress Summary Report – March 2025**
 Zones 1 and 2 Biosparge System Operations, Maintenance, and Monitoring (OM&M)
 Naval Submarine Base New London, Groton, CT
 Contract# N4008523D0073/Task Order# N4008524F6461

Dear Mr. McCloud,

KOMAN Government Solutions, LLC/Trinity (KGS/Trinity) is pleased to provide this March 2025 Monthly Progress Summary Report, detailing a summary on the Operations, Maintenance, and Monitoring (OM&M) for the biosparge system located in Zone 1 and 2, Naval Submarine Base New London, Groton, CT:

1. OM&M Activities – March 2025

- a. Performed weekly remote checks on system performance.
- b. On-site system maintenance round was completed on 04 March 2025.
 - i. KGS/Trinity tech recorded observations.
 1. Observations attached.
- c. Overall, the system is running as designed.
- d. Two VFD faults occurred on 07 March 2025 and 21 March 2025, possibly due to freezing conditions. The faults were reset.
- e. The weekly login was not performed for three weeks in March.
- f. The desired flowrate of 4.0 actual cubic feet per minute (acfm) at BSW-01 to BSW-12 was mostly achieved.
- g. The desired flowrate of 3.0 acfm at BSW-13 thru BSW-30 was mostly achieved.
- h. Light non-aqueous phase liquid (LNAPL) gauging occurred on 04 March 2025. LNAPL was detected in monitoring wells 1MW4, 1MW11, and TMW5 at estimated thicknesses of 0.36, 0.01, and 0.3 feet, respectively.
- i. LNAPL recovery at 1MW4, 1MW11, and TMW5 was conducted on 04 March 2025. A peristaltic pump was used to recover 0.75, 0.50, and 0.50 gallons of LNAPL, respectively. To date, 76.40-gallons of LNAPL have been removed at the site. A tracking summary sheet is provided as an attachment to this monthly report.

2. Submittals

- a. February 2025 Monthly Report - 04 March 2025.
- b. Draft Annual Report (2024) – 30 January 2025.

3. Next Month (April 2025) Planned O&MM Activities

- c. Next System Maintenance Round is tentatively scheduled for 09 April 2025. This will include:

1. LNAPL gauging and recovery.
- d. Address stakeholder comments on Annual Report (2024).

4. Issues and Concerns

- e. BSW-01 remains inaccessible due to a partial asphalt patch over the well cover. –
 - i. *Update 05 March 2025. Although inaccessible, air has been turned back on to this well and it is receiving the designed flowrate.*
- f. BSW-02 remains inaccessible due to a partial asphalt patch over the well cover as denoted in the June Monthly O&M Report. This well is receiving required flowrate.
- g. As indicated in the January 2023 Monthly report, 1MW9 had been destroyed by others during an excavation to repair a main water line. Recommend installing a replacement well as this well serves to monitor concentrations just before they enter the Thames River.

Distribution:

L. Bidder, NSBNL EV
P. Schauble, KGS/Trinity

Attachments:

Recorded O&M Observations 04 March 2025
LNAPL Gauging and Recovery Summary Table
Weekly System Log March 2025

Please feel free to reach out with any additional questions/concerns.

Sincerely,

KOMAN Government Solutions, LLC/Trinity

A handwritten signature in black ink, appearing to read 'Russ McCormick', with a long horizontal stroke extending to the right.

Russ McCormick, P.G. (NH, FL)
Project Manager
(978) 460-9603 - mobile

Treatment System Maintenance Log
 Biosparge Treatment System
 Non-CERCLA Zones 1, Lower Subbase, OU4
 Naval Submarine Base
 New London, Connecticut



Equipment	Suggested Maintenance	Suggested Schedule	Last Completed
Rotary Claw Compressors	Check inlet air filter, if necessary, replace	Monthly	3/4/25
	Remove dust and debris from fan and system inlet.		
Rotary Claw Compressors	Check the level of synchronizing gear oil, level should be above the middle of the sight glass. If below, the unit is leaking oil and will require repair.	Every 3 Months	2/5/25
Rotary Claw Compressors	Make sure housing is free from dust and dirt, clean if necessary	Every 6 Months	2/5/25
	Remove acoustic enclosure and clean fan cowlings, fan wheels, ventilation grilles and cooling fans		
Rotary Claw Compressors	Replace inlet air filter	Every year	N/A
	Check the inlet scree, clean if necessary		
Rotary Claw Compressors	Change synchronizing gear oil	Every 20,000 hours or 6 years, whatever comes first	N/A
Heat Exchangers	Check that the system is operating correctly	Monthly	3/4/25
	Remove dust and debris from cooling fan and heat exchanger		
Compact Filter Silencers	Remove weather hood and inspect element, clean if needed	Every 6 months	N/A
Rotameters	Remove fouling and moisture buildup in rotameter when adjusting flow becomes difficult	Inspect Monthly, as needed	3/4/25
General Inspection	Visual inspection from compressor inlet to air flow leaving the system. Inspect for leaks, system damage or any items that seem out of place. Repair as needed	Monthly	3/4/25
Electrical	Visual inspections of electrical lines for shorts, blown fuses, burnt out lights, or weather damage	Monthly	3/4/25
Biosparge and Monitoring Wells	Visually inspect biosparge and monitoring wells for broken fittings, damaged well vaults, or wells	Monthly	3/4/25

Monitoring Well Data Sheet
Biosparge Treatment System
Non-CERCLA Zones 1 & 2, Lower Subbase, OU4 Naval Submarine Base
New London, Connecticut



Date: 3/4/15
 Weather: overcast
 Operator: WAT

Location	Time	Press/Vac (psig)	VOCs (ppmv)	DO (mg/L)	Depth to LNAPL (ft TOC)	Depth to Water (ft-TOC)	LNAPL Thickness (ft)	Estimated LNAPL Well Volume* (gal)	LNAPL Volume Removed (gal)	Removal Method	Notes
MW1 -1PD1		0.00	0.0	8.60	—	5.13					
1MW1S		0.00	0.1	7.97	—	5.34					
1MW2		0.00	0.1	7.86	—	4.61					
1MW4		0.00	0.3	—	6.78	7.12	0.36	0.7	0.75	Peri	only collect data if no product is in well
1MW6		0.00	0.1	6.39	—	4.47					
1MW7		0.00	0.0	6.56	—	4.50					
1MW8		0.00	0.0	7.06	—	4.97					
1MW9											
1MW10		0.00	0.0	8.01	—	5.41					
1MW11		0.00	0.3	—	8.86	8.87	0.01		0.5	Peri	only collect data if no product is in well
TMW4		0.00	0.1	7.36	—						
TMW5		0.00	0.2	—	9.50	9.83	0.3		0.5	Peri	only collect data if no product is in well

Notes:

DO denotes dissolved oxygen.

ft denotes feet.

LNAPL denotes light non-aqueous phase liquid.

LNAPL thickness is calculated by subtracting Depth to LNAPL and Depth to Water measurements.

mg/L denotes micrograms per liter.

ppmv denotes parts per million by volume.

psig denotes pounds per square inch gauge.

TOC denotes top of casing.

VOCs denotes volatile organic compounds.

For LNAPL Volumes use the following formulas:

1-inch well - LNAPL thickness (ft) x 0.041 gal/ft

2-inch well - LNAPL thickness (ft) x 0.16 gal/ft

Treatment System Data Sheet Biosparge Treatment System
Non-CERCLA Zones 1, Lower Subbase, OU4 Naval Submarine Base
New London, Connecticut



Date: 3/4/25
 Weather: cloudy
 Operator: VNF

ID	Biosparge Well	Time	Pressure (psig)	Flow (acfm)
301	BSW1		4.2	2.0
302	BSW2		6	5.0
303	BSW3		5	4.2
304	BSW4		5.5	4.2
305	BSW5		5.0	4.4
306	BSW6		5.0	4.4
307	BSW7		5.0	4.2
308	BSW8		6.0	4.4
309	BSW9		6.0	4.0
310	BSW10		6.0	4.2
311	BSW11		4.0	4.8
312	BSW12		6.5	4.2
401	BSW13		5.0	4.6
402	BSW14		0.0	2.2
403	BSW15		5.0	3.2

ID	Biosparge Well	Time	Pressure (psig)	Flow (acfm)
404	BSW16		6.0	3.4
405	BSW17		0.0	4.8
406	BSW18		5.0	3.2
407	BSW19		5.0	3.6
408	BSW20		5.0	3.2
409	BSW21		5.0	3.2
410	BSW22		0.0	2.8
411	BSW23		5.0	3.4
412	BSW24		5.0	2.6
413	BSW25		4.0	3.2
414	BSW26		0.0	0.0
415	BSW27		5.0	3.2
416	BSW28		13.0	1.0
417	BSW29		4.0	3.2
418	BSW30		5.0	3.2

FIT-101, Main flow: 5.1 (acfm)
 FIT-201, Auxiliary flow: 3.6 (acfm)
 PI-101, Pressure: 19 (psig)
 TI-101, Temperature: 180 (°F)
 PI-102, Pressure: 18.5 (psig)
 TI-102, Temperature: 60 (°F)
 PI-201, Pressure: 19 (psig)
 TI-201, Temperature: 175 (°F)
 PI-202, Pressure: 18 (psig)
 TI-202, Temperature: 60 (°F)

BL-101, Total Runtime: 48996.4 (hours)
 BL-201, Total Runtime: 49240.7 (hours)
 TSH/TSL-801, Inside Temp.: 50 (°F)
 BL-101 VFD frequency: 50.6 (Hz)
 BL-201 VFD frequency: 50.3 (Hz)
 FIT-101, (on PLC) total flow: 139107974 (cu-ft)
 FIT-201, on PLC total flow: 157536577 (cu-ft)

PID Model-S/N: Min RAE 3000 592-910968
 Ambient ppm: 0.0 Post-cal ppm: 0.1

DO Model-S/N: Oxyguard Polaris 081063A91A0000
 Pre-cal mg/L: 8.60 Post-cal mg/L: 12.3

Table 2
LNAPL Gauging and Recovery
Summary (2021-2025)
Zones 1 and 2, Naval Submarine Base New London
Groton, Connecticut

Well ID	Well Diameter	Date	LNAPL Thickness (feet)	Estimated LNAPL Well Volume (gallon)	LNAPL Volume Removed (gallon)	Removal Method
1MW4	2-inch	6/8/2021	0.87	0.1392	0.50	Bailer
		7/8/2021	0.05	0.008	0.50	Bailer
		8/5/2021	0.02	0.0032	0.25	Bailer
		9/9/2021	ND	N/A	N/A	N/A
		10/7/2021	NM	N/A	N/A	N/A
		11/18/2021	NM	N/A	N/A	N/A
		12/16/2021	NM	N/A	N/A	N/A
		1/6/2022	NM	N/A	N/A	N/A
		2/15/2022	NM	N/A	N/A	N/A
		3/10/2022	0.1	0.016	0.25	Bailer
		4/20/2022	NM	N/A	N/A	N/A
		5/10/2022	NM	N/A	N/A	N/A
		6/13/2022	ND	N/A	N/A	N/A
		7/8/2022	NM	N/A	N/A	N/A
		8/15/2022	0.07	0.0112	1.00	N/A
		9/16/2022	0.05	0.008	1.5	Bailer
		10/7/2022	0.2	0.032	1.5	Bailer
		11/23/2022	NM	N/A	N/A	N/A
		12/8/2022	0.7	0.112	0.25	Bailer
		1/11/2023	ND	N/A	N/A	N/A
		2/15/2023	ND	N/A	N/A	N/A
		3/16/2023	ND	N/A	N/A	N/A
		4/12/2023	ND	N/A	N/A	N/A
		5/9/2023	ND	N/A	N/A	N/A
		6/2/2023	ND	N/A	N/A	N/A
		7/5/2023	0.1	0.016	0.50	Bailer
		8/10/2023	0.1	0.016	0.50	Bailer
		9/6/2023	ND	N/A	N/A	N/A
		10/12/2023	ND	N/A	N/A	N/A

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1MW4	2-inch	11/20/2023	ND	N/A	N/A	N/A
		12/7/2023	ND	N/A	N/A	N/A
		1/11/2024	ND	N/A	N/A	N/A
		2/8/2024	ND	N/A	N/A	N/A
		3/19/2024	ND	N/A	N/A	N/A
		4/11/2024	ND	N/A	N/A	N/A
		5/16/2024	ND	N/A	N/A	N/A
		11/15/2024	ND	N/A	N/A	N/A
		12/3/2024	0.1	0.016	0.50	Peristaltic
		1/5/2025	1	0.16	0.50	Peristaltic
		2/5/2025	0.2	0.032	0.50	Peristaltic
		3/4/2025	0.36	0.0576	0.75	Peristaltic
1MW4 Removed Total CY2021					1.25	
1MW4 Removed Total CY2022					4.50	
1MW4 Removed Total CY2023					1.00	
1MW4 Removed Total CY2024					0.50	
1MW4 Removed Total CY2025					1.75	
1MW4 Removed Subtotal					9.00	
1MW10	2-inch	11/23/2022	0	0	0.5	Bailer
		1/11/2023	ND	N/A	N/A	N/A
		2/15/2023	ND	N/A	N/A	N/A
		3/16/2023	NM	N/A	N/A	N/A
		4/12/2023	NM	N/A	N/A	N/A
		5/9/2023	NM	N/A	N/A	N/A
		6/2/2023	ND	N/A	N/A	N/A
		7/5/2023	ND	N/A	N/A	N/A
		8/10/2023	ND	N/A	N/A	N/A
		9/6/2023	ND	N/A	N/A	N/A
		10/12/2023	ND	N/A	N/A	N/A
		11/20/2023	ND	N/A	N/A	N/A
		12/7/2023	ND	N/A	N/A	N/A

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Groton, Connecticut

Well ID	Well Diameter	Date	LNAPL Thickness (feet)	Estimated LNAPL Well Volume (gallon)	LNAPL Volume Removed (gallon)	Removal Method
1MW10	2-inch	1/11/2024	ND	N/A	N/A	N/A
		2/8/2024	ND	N/A	N/A	N/A
		3/19/2024	ND	N/A	N/A	N/A
		4/11/2024	ND	N/A	N/A	N/A
		5/16/2024	ND	N/A	N/A	N/A
		11/15/2024	ND	N/A	N/A	N/A
		12/3/2024	ND	N/A	N/A	N/A
		1/3/2025	ND	N/A	N/A	N/A
		2/5/2025	ND	N/A	N/A	N/A
		3/4/2025	ND	N/A	N/A	N/A
1MW10 Removed Total CY2022					0.50	
1MW10 Removed Total CY2023					0.00	
1MW10 Removed Total CY2024					0.00	
1MW10 Removed Total CY2025					0.00	
1MW10 Removed Subtotal					0.50	
1MW11	2-inch	6/8/2021	ND	N/A	N/A	N/A
		7/8/2021	ND	N/A	N/A	N/A
		8/5/2021	ND	N/A	N/A	N/A
		9/9/2021	ND	N/A	N/A	N/A
		10/7/2021	0.16	0.0256	1.25	Bailer
		11/18/2021	0.56	0.0896	1.5	Bailer
		12/16/2021	0.15	0.024	1	Bailer
		1/6/2022	ND	N/A	N/A	N/A
		2/15/2022	0.02	0.0032	1.25	Bailer
		3/10/2022	0.5	0.08	0.5	Bailer
		4/20/2022	0.1	0.016	0.5	Bailer
		5/10/2022	Only on probe	N/A	0.25	Bailer
		6/13/2022	0.01	0.0016	0.25	Bailer
		7/8/2022	ND	N/A	N/A	N/A
		8/15/2022	ND	N/A	N/A	N/A
		9/16/2022	0.21	0.0336	2	Bailer

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Groton, Connecticut

Well ID	Well Diameter	Date	LNAPL Thickness (feet)	Estimated LNAPL Well Volume (gallon)	LNAPL Volume Removed (gallon)	Removal Method
1MW11	2-inch	10/7/2022	1.04	0.1664	1	Bailer
		11/23/2022	0	0	0.5	Bailer
		12/8/2022	0.03	0.0048	0.25	Bailer
		1/11/2023	2.15	0.344	2	Bailer
		2/15/2023	ND	N/A	N/A	N/A
		3/16/2023	ND	N/A	N/A	N/A
		4/12/2023	ND	N/A	N/A	N/A
		5/9/2023	ND	N/A	N/A	N/A
		6/2/2023	ND	N/A	N/A	N/A
		7/5/2023	ND	N/A	N/A	N/A
		8/10/2023	ND	N/A	N/A	N/A
		9/6/2023	0.9	0.144	1	Bailer
		10/12/2023	0.2	0.032	1	Bailer
		11/20/2023	0.5	0.08	1	Bailer
		12/7/2023	0.3	0.048	0.5	Peristaltic
		1/11/2024	0.1	0.016	0.5	Peristaltic
		2/8/2024	ND	N/A	N/A	N/A
		3/19/2024	ND	N/A	N/A	N/A
		4/11/2024	0.5	0.08	0.5	Peristaltic
		5/16/2024	0.2	0.032	0.5	Peristaltic
		11/15/2024	0.01	0.0016	0.15	Peristaltic
		12/3/2024	0.1	0.016	0.5	Peristaltic
		1/3/2025	0.1	0.016	0.5	Peristaltic
		2/5/2025	0.05	0.008	0.25	Peristaltic
		3/4/2025	0.01	0.0016	0.5	Peristaltic
1MW11 Removed Total CY2021					3.75	
1MW11 Removed Total CY2022					6.50	
1MW11 Removed Total CY2023					5.50	
1MW11 Removed Total CY2024					2.15	
1MW11 Removed Total CY2025					1.25	
1MW11 Removed Subtotal					19.15	

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Zones 1 and 2, Naval Submarine Base New London
Groton, Connecticut

Well ID	Well Diameter	Date	LNAPL Thickness (feet)	Estimated LNAPL Well Volume (gallon)	LNAPL Volume Removed (gallon)	Removal Method
TMW5	1-inch	6/8/2021	0.9	0.0369	0.5	Bailer
		7/8/2021	1.09	0.04469	0.75	Bailer
		8/5/2021	1.48	0.06068	0.75	Bailer
		9/9/2021	1.34	0.05494	2	Bailer
		10/7/2021	0.71	0.02911	0.75	Bailer
		11/18/2021	2.09	0.08569	0.5	Bailer
		12/16/2021	1.5	0.0615	1.5	Bailer
		1/6/2022	1.39	0.05699	1.5	Bailer
		2/15/2022	0.04	0.00164	0.25	Bailer
		3/10/2022	0.67	0.02747	1.25	Bailer
		4/20/2022	3.00	0.123	1.5	Bailer
		5/10/2022	1.85	0.07585	0.75	Bailer
		6/13/2022	0.9	0.0369	0.75	Bailer
		7/8/2022	0.48	0.01968	1.25	Bailer
		8/15/2022	0.52	0.02132	1.5	Bailer
		9/16/2022	NM	N/A	N/A	N/A
		10/7/2022	2.7	0.1107	1	Bailer
		11/23/2022	NM	N/A	N/A	N/A
		12/8/2022	2.85	0.11685	1.5	Bailer
		1/11/2023	4.1	0.1681	0.25	Bailer
		2/15/2023	NM	N/A	N/A	N/A
		3/16/2023	6	0.246	2.0	Peristaltic
		4/12/2023	3.8	0.1558	1.5	Bailer/Peristaltic
		5/16/2023	NM	N/A	1.0	Bailer
		6/2/2023	3.7	0.1517	1.5	Bailer
		7/5/2023	2	0.082	1.5	Bailer
		8/10/2023	1	0.041	1.5	Bailer
		9/6/2023	3.8	0.1558	1.0	Bailer
		10/12/2023	2.1	0.0861	1.5	Bailer
		11/2/2023	2.4	0.0984	N/A*	N/A
		12/7/2023	2.4	0.0984	1.5	Peristaltic

Table 2
LNAPL Gauging and Recovery
Summary (2021-2025)
Zones 1 and 2, Naval Submarine Base New London
Groton, Connecticut

Well ID	Well Diameter	Date	LNAPL Thickness (feet)	Estimated LNAPL Well Volume (gallon)	LNAPL Volume Removed (gallon)	Removal Method
TMW5	1-inch	1/11/2024	5	0.205	2.0	Peristaltic
		2/28/2024	9.3	0.3813	2.0	Peristaltic
		3/19/2024	7	0.287	2.0	Peristaltic
		4/11/2024	4	0.164	1.5	Peristaltic
		5/16/2024	4	0.164	1.5	Peristaltic
		11/15/2024	0.3	0.0123	1.0	Peristaltic
		12/3/2024	3.7	0.1517	2.0	Peristaltic
		1/3/2025	0.7	0.0287	2.0	Peristaltic
		2/5/2025	1.9	0.0779	2.0	Peristaltic
		3/4/2025	0.3	0.0123	0.5	Peristaltic
TMW5 Removed Total CY2021					6.75	
TMW5 Removed Total CY2022					11.25	
TMW5 Removed Total CY2023					13.25	
TMW5 Removed Total CY2024					12.00	
TMW5 Removed Total CY2025					4.50	
TMW5 Removed Subtotal					47.75	
Cumulative Removed Total CY2021					11.75	
Cumulative Removed Total CY2022					22.75	
Cumulative Removed Total CY2023					19.75	
Cumulative Removed Total CY2024					14.65	
Cumulative Removed Total CY2025					7.50	
Cumulative Total Removed					76.40	

Notes:

* LNAPL recovery did not occur. See monthly report.

LNAPL denotes light non-aqueous phase liquid.

N/A denotes not applicable.

ND denotes LNAPL was not detected.

NM denotes not measured due in access issues with well.

Only wells with historical LNAPL are shown on the table. Wells that are not shown have not had LNAPL detected.

NSBNL Biosparge
Weekly System Check



Date	Time	Operator	HE-101 ON/OFF	HE-201 ON/OFF	BL-101 ON/OFF	BL-101 VFD (Hz)	BL-201 ON/OFF	BL-201 VFD (Hz)	FIT-101 (scfm)	FIT-201 (scfm)	FIT-101 Total Flow (cuFT)	FIT-201 Total Flow (cuFT)	Alarms / Faults Active	Combined Injection Rate (scfm)	Total Combined Flow Volume (FIT-101 + FIT- 201) (cuFT)	Weekly Injection Flow Vol (cuFT)	Comments
3/6/2025	630	MS	ON	ON	ON	51	ON	51	51	55	139244297	157681222		106	296,925,519		